

# arithmetic operations

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## Introduction

In this laboratory you will write your own assembly programs to perform additions and subtractions on the CT Board.

## Learning Objectives

- You can apply addition and subtraction operations in assembly programs.
- You understand the implications of these operations on the processor flags.
- You can implement additions, which exceed the word size of the processor.

## Task 1: Sum and difference

Today's laboratory is structured into three projects. To build each project with Makefiles, you need to change into the projects directory, or use `make -C <path/to/project> <rule-name>`.

You shall write a program that adds and subtracts two 8 bit values. To be able to observe the carry and overflow flags, the two 8 bit values will be shifted to the left by 24 bits to occupy the most significant eight bits of the register.

Open the given project frame `sum_diff` and expand it, so that one 8 bit wide value is read from the DIP-switches `SW[15..8]` (operand A) and another from `SW[7..0]` (operand B). Both operands shall be expanded to 32 bit, as mentioned in the introduction. Display the most significant byte of the sum on `LED[7..0]` and the most significant byte of the difference on `LED[23..16]`.

The instruction `lslls` (i.e. `lslls r1, r1, #24`) shifts the content of register `r1` to the left by 24 bits. The right part gets filled with zeroes. You will learn about the shift instructions in a future lesson.

Additionally you shall display the flags set by these two operations on the LEDs. The flags from the addition shall be displayed on `LED[15..12]` and the ones from the subtraction on `LED[31..28]` (See the figure below).

- To read the processor flags, you can use the `mrs` instruction. i.e.: `mrs r1, apsr` copies the content of the application program status register to register r1.
- The four flag bits are positioned from bit 31 to bit 28. To display these on the LEDs, you need to shift those bits to the right by 24 bits. Use the `lsrs` instruction. i.e.: `lsrs r1, r1, #24` shifts the content of r1 to the right by 24 bits.

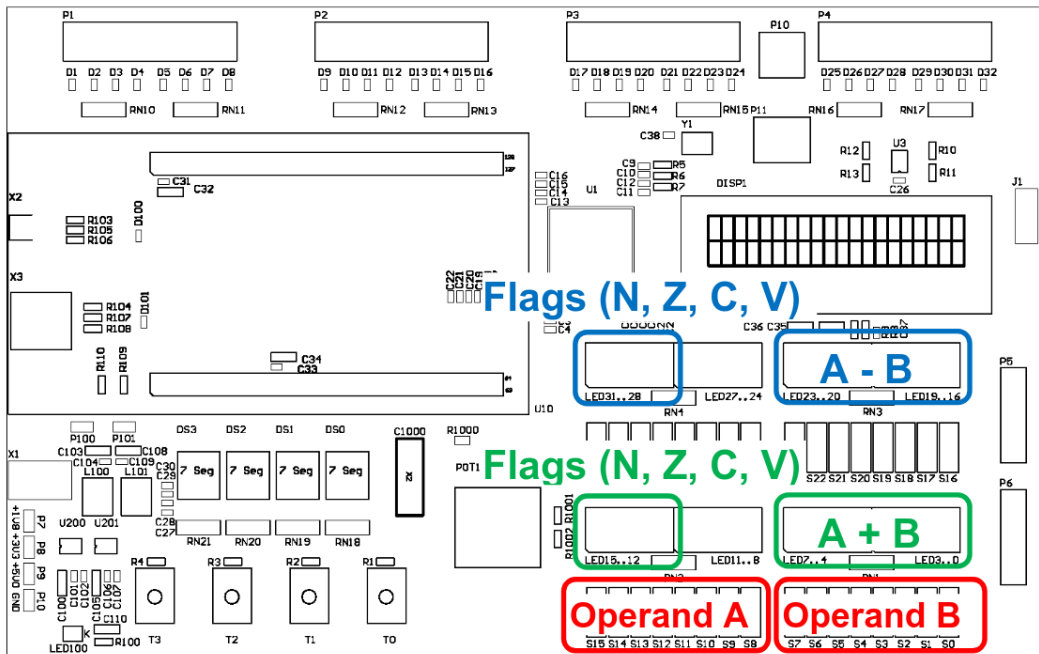


Figure 1: In and output of values for the sum and difference

Before writing your program, calculate the expected results based on the following table, i.e. sum, difference and flags.

### Addition

| A    | B    | Result (A+B) | N | Z | C | V |
|------|------|--------------|---|---|---|---|
| 0x82 | 0x12 |              |   |   |   |   |
| 0x34 | 0x72 |              |   |   |   |   |
| 0xC2 | 0x87 |              |   |   |   |   |
| 0xA3 | 0x62 |              |   |   |   |   |
| 0x67 | 0x99 |              |   |   |   |   |

### Subtraction

| A    | B    | Result (A-B) | N | Z | C | V |
|------|------|--------------|---|---|---|---|
| 0xEE | 0x12 |              |   |   |   |   |
| 0x8E | 0x72 |              |   |   |   |   |
| 0x79 | 0x87 |              |   |   |   |   |
| 0x9E | 0x62 |              |   |   |   |   |
| 0x67 | 0x99 |              |   |   |   |   |

Note for subtraction: **C** = '0' means 'borrow', while **C** = '1' means, that 'borrow' is not necessary.

Once the tables are completed write the program and compare its output against the tables.

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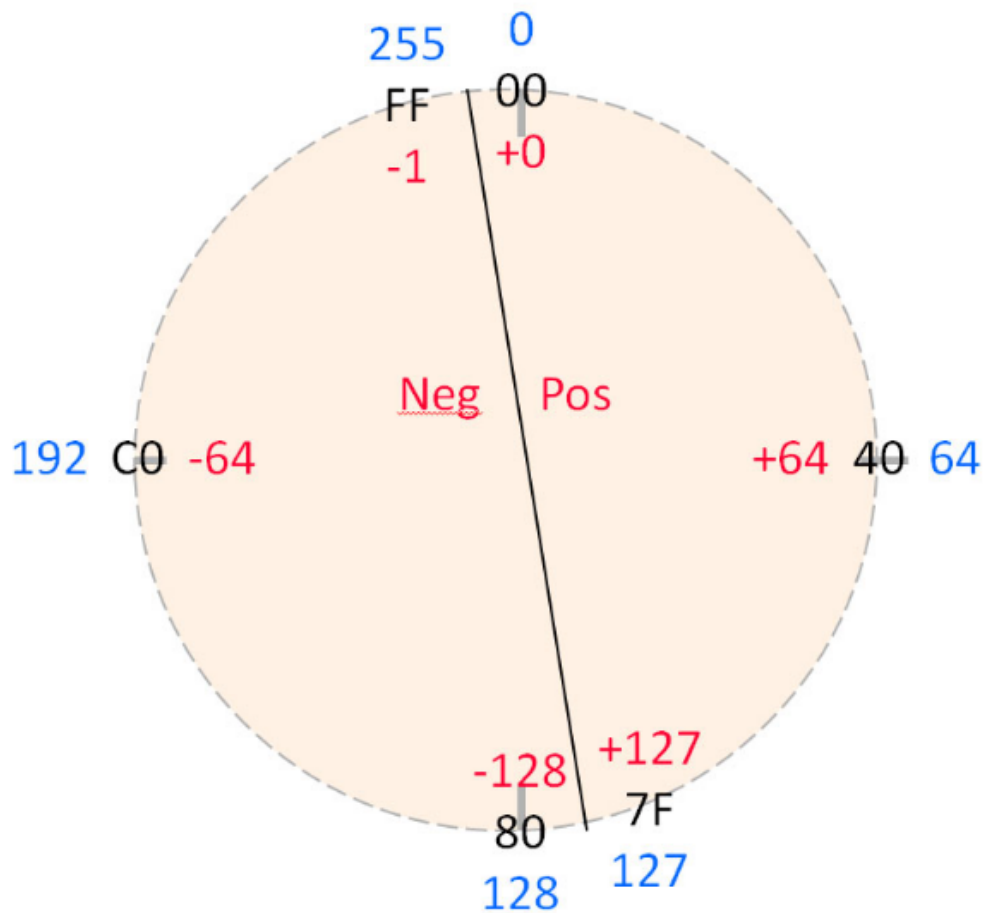


Figure 2: number circle

## Task 2: 64 Bit Addition

Extend the provided project `add64` with a 64 bit summation variable (unsigned interpretation). Every time you press button `T0`, the program shall read a 32 bit input value from the DIP-switches and shall add the read value to the summation variable. The sum shall be displayed continuously on the LCD display (see the image below).

Use the binary interface of the LCD display (see page LCD Binary Interface on the [CT wiki](#)).

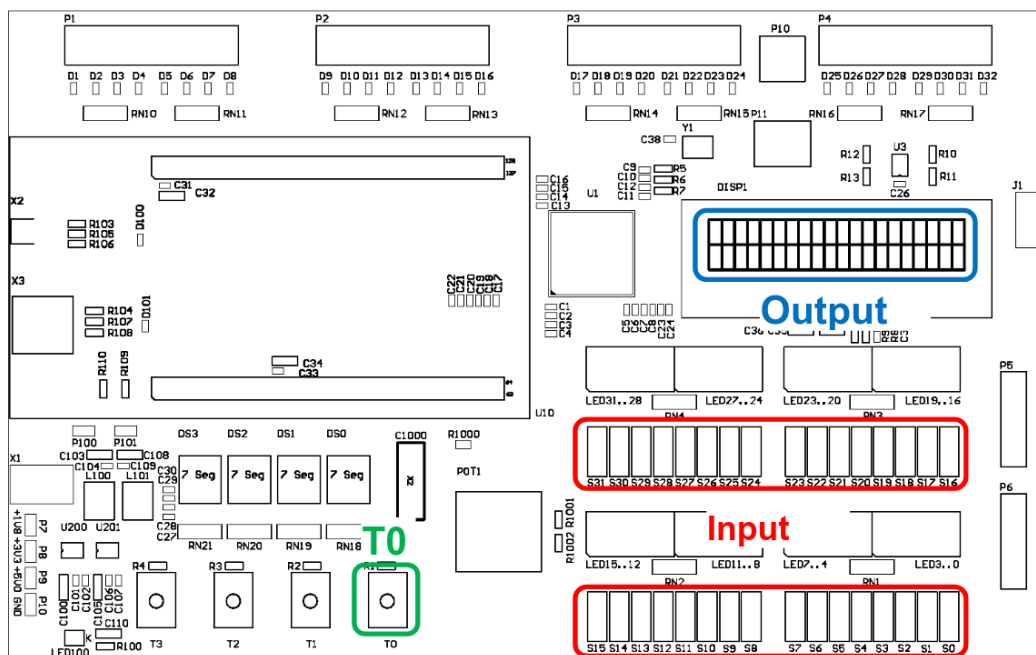


Figure 3: In- and output for the 64 bit addition

Verify the results of your program, especially the overflow from one word to the other.

### Task 3 (optional): Arithmetic Commands

Assemble, link and load the given project arith\_operations with the debugger. Execute the program step by step. Watch and comprehend the changes in the registers and flags.

### Grading

The working programs have to be presented to the lecturer. The student has to understand the solution / source code and has to be able to explain it to the lecturer.

| Criteria                                                                 | Weight |
|--------------------------------------------------------------------------|--------|
| The program meets the requirements described in <a href="#">Task 1</a> . | 2 / 4  |
| The program meets the requirements described in <a href="#">Task 2</a> . | 2 / 4  |